Supporting Information

for: Using LiDAR to quantify, map, and conserve late-successional and old-growth forest in Maine, USA: Comment (A. R. Weiskittel).

Appendix S1. Methods

Reproduction. The published random forest was refit from the archived 463-plot training data (8 LiDAR canopy metrics) with mtry=2 and 500 trees, reproducing the binary out-of-bag accuracy (94.2% vs 94.1%) and the top predictor. The model was applied to the archived per-hectare LiDAR metrics for the full area of interest (4,282,675 ha) to reproduce the wall-to-wall classification, and a 20-seed ensemble bounded the area estimate and the cross-validation agreement (Table S3).

Cross-map comparison. Three wall-to-wall classifications were placed on a common 100 m grid: the reproduced Hagan classifier; a logistic model of the FIA field class on Potapov (GEDI-calibrated) canopy height (calibrated to the FIA plot prevalence); and the USFS TreeMap 2022 imputation of FIA structural class. Pairwise Cohen kappa and Jaccard, three-way concordance, and the overlap of top-priority protected sets were computed.

Accuracy on the training plots. Cross-validated AUC (repeated stratified five-fold, 40 repeats) was estimated for three predictor sets (8 LiDAR metrics; canopy height only; ground structure) against three targets, under random forest and logistic regression (Table S1). The capacity of the LiDAR metrics to predict the ground structural attributes that define old growth was assessed by cross-validated R-squared (Table S6).

FIA design-based estimation. Design-based older-forest area and 2003-2024 trend were estimated with rFIA (post-stratified, with FIA estimation-unit areas) over all Maine inventory years, for stand-age and large-tree basal-area domains, statewide and in the northern timberland units. Sensitivity to the large-tree threshold is in Table S2.

Hex-scale summary. Each method's any-LSOG was aggregated to 8 km hexagons (n = 984) over the area of interest; the cross-method disagreement (maximum minus minimum any-LSOG fraction per hexagon) had mean 0.22 and median 0.19 (Fig. S1).

Data and code. All scripts and derived products are archived at Zenodo (concept DOI 10.5281/zenodo.20614496). FIA data are from the USDA FIA DataMart; Hagan training data and code from Zenodo 10.5281/zenodo.19696494.

Table S1. AUC by classifier and predictor set

Table S1. Cross-validated AUC (mean and 95% interval) by classifier (random forest, logistic regression), target, and predictor set. Canopy-height-only is weakest for old growth under both classifiers.

classifier	target	predictors	auc	lo	hi
glm	any-LSOG	LIDAR8	0.99	0.988	0.991
glm	any-LSOG	HEIGHT	0.989	0.989	0.99
glm	any-LSOG	STRUCT	0.968	0.964	0.971
glm	LS+OG	LIDAR8	0.983	0.981	0.985
glm	LS+OG	HEIGHT	0.985	0.982	0.986
glm	LS+OG	STRUCT	0.977	0.972	0.98
glm	OG	LIDAR8	0.967	0.921	0.991
glm	OG	HEIGHT	0.912	0.893	0.924
glm	OG	STRUCT	0.973	0.966	0.978
rf	any-LSOG	LIDAR8	0.99	0.988	0.992
rf	any-LSOG	HEIGHT	0.984	0.982	0.987
rf	any-LSOG	STRUCT	0.965	0.961	0.968
rf	LS+OG	LIDAR8	0.987	0.982	0.989
rf	LS+OG	HEIGHT	0.977	0.974	0.979
rf	LS+OG	STRUCT	0.969	0.966	0.972
rf	OG	LIDAR8	0.967	0.952	0.979
rf	OG	HEIGHT	0.913	0.868	0.933
rf	OG	STRUCT	0.975	0.97	0.98

Table S2. FIA design-based sensitivity to the large-tree threshold

Table S2. Design-based older-forest area (large-tree basal area domain) at three thresholds, with 95% CI and 2003-2024 trend. The increasing trend is robust to the threshold.

domain	yr	2024 %	lo	hi	trend %/yr	slope lo	slope hi
Large-tree BA >= 20 ft2/ac	2024	19.516	18.151	20.88	0.161	0.136	0.186
Large-tree BA >= 30 ft2/ac	2024	12.498	11.349	13.647	0.168	0.151	0.185
Large-tree BA >= 40 ft2/ac	2024	8.179	7.217	9.14	0.114	0.096	0.133

Table S2b. FIA older forest by ownership (private commercial vs public)

Table S2b. Design-based older-forest area by ownership group with 95% CI and 2003-2024 trend. Older forest is rising on private commercial timberland as well as public land.

domain	owner	2024 %	lo	hi	trend %/yr	slope lo	slope hi
Stand age >= 120 yr	private	3.313	2.657	3.969	0.031	0.019	0.044
Stand age >= 120 yr	public	10.797	7.124	14.471	0.391	0.326	0.456
Large-tree BA >= 30	private	11.543	10.377	12.709	0.15	0.134	0.165
Large-tree BA >= 30	public	22.943	17.633	28.253	0.21	0.111	0.309

Table S3. Random forest stochasticity (20-seed ensemble)

Table S3. Mean, SD, and range across 20 random forest seeds for reproduced accuracy, wall-to-wall area, and cross-validation agreement.

metric	mean	sd	min	max
oob_binary_acc	94.05	0.317	93.737	95.032
ww_any_pct	21.591	0.22	21.15	22.1
ww_lsogl_pct	4.235	0.063	4.115	4.339
ww_ogl_pct	0.945	0.032	0.88	1.007
ww_lsogl_ha	181,317	2,707	176,207	185,787
kappa_any	0.128	0.004	0.122	0.139
kappa_lsog	0.064	0.011	0.049	0.079

Table S4. Cross-method disagreement by owner class

Table S4. Of the LSOG flagged by any method, the share that is contested (flagged by one or two methods, not all) by owner class. Disagreement is pervasive across ownership.

owner code	flagged ha	all-agree ha	contested ha	% contested
3	364,952	1,624	363,328	99.6
0	344,603	1,448	343,155	99.6
4	314,352	899	313,453	99.7
7	147,168	989	146,179	99.3
1	102,091	125	101,966	99.9
6	42,120	594	41,526	98.6
8	16,199	48	16,151	99.7
2	16,046	135	15,911	99.2
5	719	0	719	100
	530	2	528	99.6

Table S5. TreeMap LSOG time series

Table S5. TreeMap-imputed LSOG area over 2016-2022 (native 30 m), Maine unorganized townships.

year	any-LSOG %	LS+OG %	known ha
2016	6.452	0.217	3,765,076
2020	7.902	0.608	3,888,375
2022	7.806	0.594	3,876,445

Table S6. LiDAR prediction of structure, and old-growth discriminators

Table S6a. Cross-validated R-squared for predicting ground structural attributes from the eight LiDAR metrics. Dead wood (CWD, snags) is poorly predicted.

structural attribute	CV R-squared from LiDAR	lo	hi
basal_GE_40cmDBH	0.681	0.669	0.7
No_treesGE_40cmdbh	0.586	0.578	0.597
CWD_vol	0.201	0.175	0.23
sum_dead_basalarea	0.241	0.222	0.264
CV_livedbh	0.514	0.502	0.526

Table S6b. Old-growth discriminators (random forest variable importance) across LiDAR and ground-structure variables.

variable	set	importance (MDA)
sum_live_basalarea	ground structure	undefined
cano_cover_15	LiDAR canopy	undefined
basal_GE_40cmDBH	ground structure	undefined
percentile_95th	LiDAR canopy	undefined
CV_livedbh	ground structure	undefined
QMD_live	ground structure	undefined
mean_cano_ht	LiDAR canopy	undefined
sum_dead_basalarea	ground structure	undefined
cano_cover_2	LiDAR canopy	undefined
max_cano_ht	LiDAR canopy	undefined
No_treesGE_40cmdbh	ground structure	undefined
prop_basal_GE_40cmDBH	ground structure	undefined
top_rugosity	LiDAR canopy	undefined
cano_cover_6	LiDAR canopy	undefined
rumple	LiDAR canopy	undefined
CWD_vol	ground structure	undefined

Table S7. Reproduced random forest out-of-bag confusion matrix

Table S7. Out-of-bag confusion matrix (counts) for the reproduced airborne-LiDAR random forest. Rows are field-assigned classes, columns predicted. True old growth is recovered only about a quarter of the time (Fig. 1 of the Comment).

True	LS	Not LS	Old-growth	Trans LS
LS	68	0	1	10
Not LS	0	268	0	14
Old-growth	9	0	4	4
Trans LS	10	15	1	59

Table S8. Sensitivity of the classification to the probability cutoff

Table S8. As the random forest probability cutoff for calling LSOG varies, the fraction of labelled plots called LSOG, sensitivity, specificity, overall accuracy, and the true-skill statistic (TSS) all shift (Fig. 2 of the Comment).

cutoff	frac called LSOG	sensitivity	specificity	accuracy	TSS
0.1	0.944	1	0.329	0.886	0.329
0.15	0.927	0.997	0.418	0.898	0.415
0.2	0.909	0.992	0.494	0.907	0.486
0.25	0.888	0.99	0.608	0.924	0.597
0.3	0.868	0.984	0.696	0.935	0.681
0.35	0.862	0.982	0.722	0.937	0.703
0.4	0.849	0.971	0.747	0.933	0.718
0.45	0.836	0.966	0.797	0.937	0.764
0.5	0.823	0.958	0.835	0.937	0.794
0.55	0.812	0.948	0.848	0.931	0.796
0.6	0.808	0.943	0.848	0.927	0.791
0.65	0.786	0.93	0.911	0.927	0.841
0.7	0.778	0.922	0.924	0.922	0.846
0.75	0.769	0.914	0.937	0.918	0.851
0.8	0.76	0.904	0.937	0.909	0.84
0.85	0.749	0.893	0.949	0.903	0.843
0.9	0.728	0.875	0.987	0.894	0.862

Table S9. Independent third-party old-growth comparison (Pelz et al. 2023)

Table S9. Plot-level agreement (Cohen's kappa) between the field-structure classification used here and the U.S. Forest Service national old-growth inventory (Pelz et al. 2023), evaluated on the FIA plots where both apply: National Forest System land in Maine, New Hampshire, New York, and Vermont (n about 925, concentrated in NH and VT because Maine holds little federal forest). Agreement is weak even between two independently authored operationalizations of old forest, corroborating the cross-map disagreement documented in the Comment.

Comparison	Cohen's kappa
v5.1 old growth vs Pelz old growth	0.04
v5.1 late-successional + old growth vs Pelz old growth	0.25
v5.1 any-LSOG vs Pelz old growth	0.12

Table S10. FIA older-forest trend clipped to the published map area of interest

Table S10. FIA design-based older-forest area within Hagan et al.'s exact area-of-interest polygon (the unorganized townships, about 4.28 million ha), by rFIA spatial estimation (polys), 2003 to 2024. Within the map's own footprint older forest is increasing on every measure, with all weighted-regression slopes positive and 95% intervals excluding zero. This confirms the statewide trend is not masking a within-UT decline.

Ground criterion	2003 (%)	2024 (%)	Trend %/yr [95% CI]
Stand age >= 120 yr	3.1	4.0	+0.058 [0.043, 0.073]
Stand age >= 100 yr	9.4	13.2	+0.199 [0.149, 0.249]
Large-tree BA >= 30 ft-sq/ac	9.1	9.7	+0.052 [0.036, 0.069]

Figure S1. Hex-scale cross-map disagreement

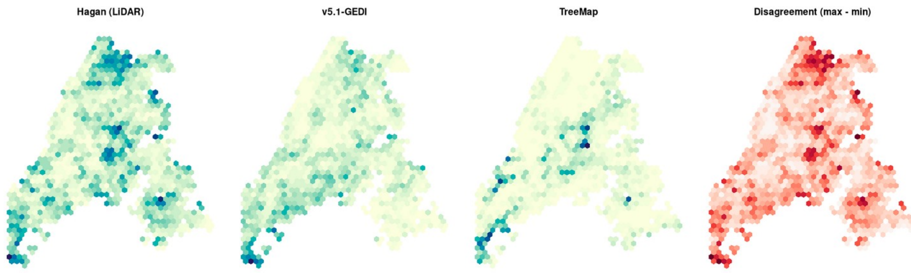

Figure S1. Any-LSOG fraction by method aggregated to 8 km hexagons over the area of interest, and the cross-method disagreement (max minus min). Disagreement concentrates in the northern and central townships.

Figure S2. Maine study-area ensemble (agreement and uncertainty)

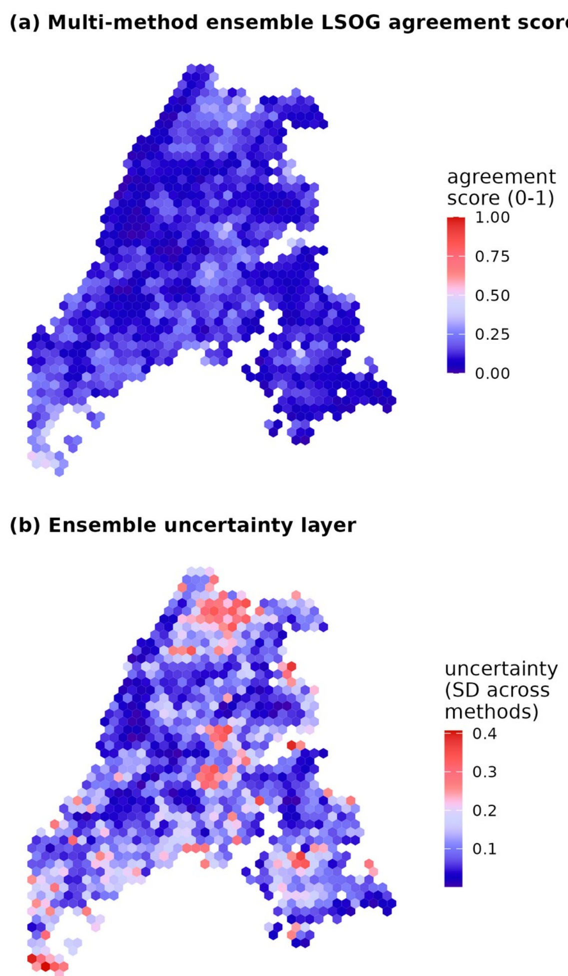

Figure S2. Prototype multi-method ensemble for the Maine study area, aggregated to 8 km hexagons: (a) agreement score (unweighted mean of three method memberships) and (b) uncertainty (standard deviation). The companion New England version is Fig. 7 of the Comment.

Figure S4. Older forest over time under different definitions and thresholds

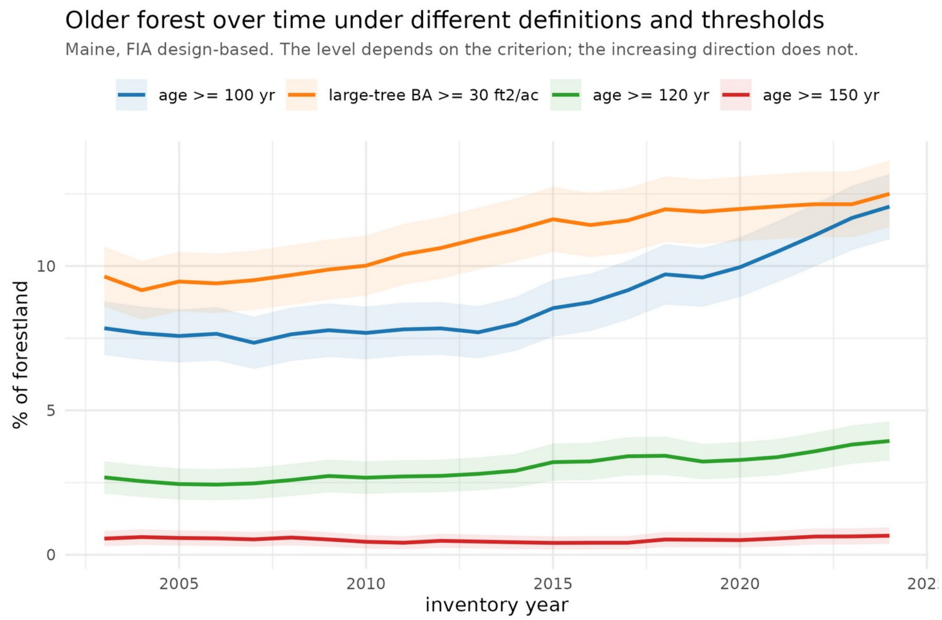

Figure S4. Maine older forest over time (FIA design-based) under four ground definitions: stand age >= 100, >= 120, and >= 150 yr, and large-tree basal area >= 30 ft-sq/ac, each with a 95% band. The level depends on the criterion, but every definition increases over 2003-2024.

Figure S3. Maine ensemble bivariate (agreement by uncertainty)

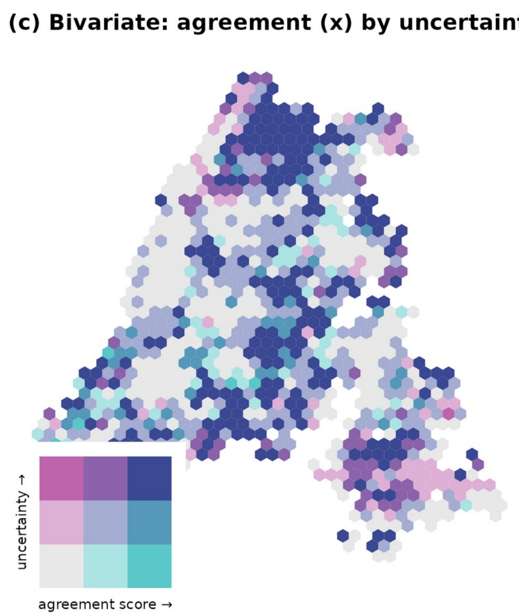

Figure S3. Bivariate map of the prototype ensemble agreement score (horizontal) by uncertainty (vertical) for the Maine study area, tercile breaks. The defensible acquisition targets are the high-agreement, low-uncertainty hexes; most of the landscape carries high uncertainty.

Figure S5. Why LSOG is where it is: terrain and disturbance

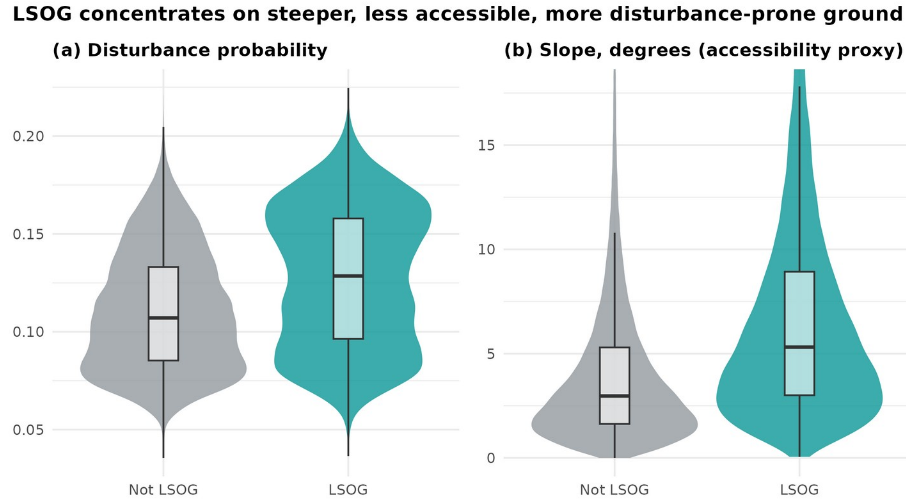

Figure S5. Distributions of (a) modeled disturbance probability and (b) slope (degrees, an accessibility proxy) for LSOG versus non-LSOG pixels over the Maine AOI (n = 3.8 million). LSOG concentrates on steeper, less accessible, more disturbance-prone ground; the LSOG share rises from 12% to 37% across slope terciles and from 17% to 35% across disturbance terciles. Aaron's CONUS harvest-probability model does not cover the Maine unorganized townships, so slope stands in for harvestability.

Figure S6. A Landsat continuity layer for true-LSOG mapping

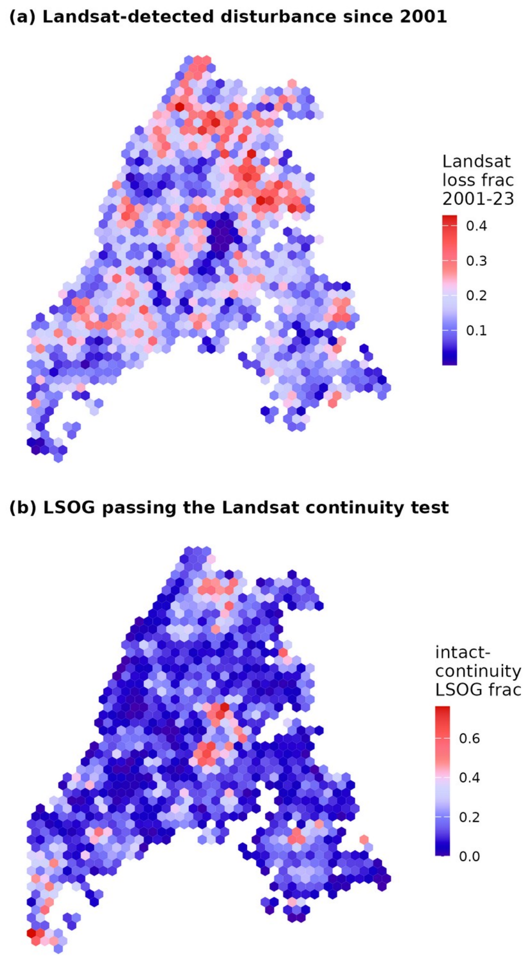

Figure S6. The temporal-continuity axis made mappable from Landsat (Hansen Global Forest Change, 2001-2023), 8 km hexagons: (a) fraction of each hex with stand-replacing canopy loss since 2001; (b) fraction of the canopy-mapped LSOG that is intact over the record. Of canopy-mapped LSOG, 24% shows stand-replacing disturbance since 2001 (failing strict continuity) and 76% is intact; partial harvests and pre-2001 entry (legacy skid trails) are not captured, so this is a lenient test.